

Acute and subacute toxicity of tyramine, spermidine, spermine, putrescine and cadaverine in rats.

The acute and subacute toxicity of five biogenic amines-tyramine, spermidine, spermine, putrescine and cadaverine-were examined in Wistar rats. Tyramine and cadaverine had a low acute oral toxicity of more than 2000 mg/kg body weight. Putrescine had an acute oral toxicity of 2000 mg/kg body weight and spermidine and spermine each of 600 mg/kg body weight. All amines investigated caused a dose-related decrease in blood pressure after intravenous administration, except for tyramine, where an increase was found. In 6-wk studies the biogenic amines were administered in the diet to groups of 10 male and 10 female rats. Tyramine and cadaverine were given at levels of 0, 200, 2000 or 10,000 ppm, spermine and putrescine at levels of 0, 200, 2000 or 5000 ppm and spermidine at levels of 0, 20, 200 or 500/1000 ppm in the first study and at levels of 0 or 10,000 ppm in a second study. Spermine was the most toxic. The high dose level showed a great number of changes, such as emaciation, aggressiveness, convulsions and paralysis of the hind legs. Growth, food intake and water intake were considerably decreased. Slight anaemia (males) and changes in plasma clinical chemistry occurred. The relative weights of the thyroid, adrenals, spleen and heart were increased and that of the liver decreased. Impaired kidney function, together with renal histopathological changes and changes in plasma electrolytes and urea, occurred with spermine. Histopathological examinations also revealed decreased glycogen content in the liver, reduction of spermatogenesis, severe depletion of splenic white pulp, acute involution of the thymus and moderate myocardial degeneration in the heart. Myocardial degeneration was also seen in one mid-dose male. Adverse effects were also observed in the top dose groups of all other amines. Decreased body weights associated with diminished food intake were generally seen. Slight increases in packed cell volume, haemoglobin concentration and thrombocytes occurred with cadaverine. With spermidine, decreased plasma creatinine, calcium and inorganic phosphate were observed and decreased potassium levels with cadaverine. The no-observed-adverse-effect level was 2000 ppm (180 mg/kg body weight/day) for tyramine, cadaverine and putrescine, 1000 ppm (83 mg/kg body weight/day) for spermidine and 200 ppm (19 mg/kg body weight/day) for

spermine.